

Koa Shen

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Hiring Team

Parker Hannifin

Re: Summer 2027 Engineering Internship

Dear Parker Hannifin Hiring Team,

I am a mechanical engineering student at UC Santa Barbara applying for a Summer 2027 engineering internship. Parker Hannifin's work developing motion and control technologies for demanding applications stands out to me because it calls for the kind of practical, systems-minded engineering I enjoy: designing mechanisms, understanding how they behave, and validating that they perform reliably in the real world.

At SLAC National Accelerator Laboratory, I built an open-source Python and Drake digital-twin framework for tightly packaged motion assemblies. The framework brings CAD geometry and kinematics together for live collision checks, helping engineers identify interference risks and make informed design changes before hardware is operated. I have also redesigned components of an X-ray optics assembly and performed structural checks for stages and mounts protecting detectors valued at more than \$100,000. This experience has made me attentive to clearances, load cases, manufacturability, and the consequences of failure.

Outside that work, I designed and bench-tested a compact 15:1 cycloidal reducer packaged within a NEMA 17 motor footprint, then wrote C++ firmware using encoder feedback to correct position error and detect missed steps. With UCSB Gaucho Racing, I lead fabrication work and CNC-machine suspension components, holding bearing interfaces within 0.0005 inches. These projects have taught me to carry designs through analysis, fabrication, assembly, and testing, then refine them based on what the hardware reveals.

I would bring Parker Hannifin a strong foundation in mechanical design, motion systems, simulation, embedded controls, and hands-on manufacturing, along with the curiosity to learn from experienced engineers. I would be excited to contribute to products that help customers move, control, and protect critical systems.

Thank you for your consideration.

Sincerely,

Koa Shen